Is there an association between dietary fiber intake, screen time, and physical activity among students aged 14 to 17 years?—a cross-sectional study

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Resumo (Abstract)

Background Concerns about the impact of screen time, unbalanced diet, and declining physical activity (PA) on health are increasing. Therefore, the aim of the present study was to examine the association between screen time, PA and dietary behavior, with a special consideration of fiber intake in adolescents. Methods The study is based on cross-sectional data including 410 Austrian adolescents (54.6% girls) with an average age of 15.7±1.0 years. Daily fiber intake was determined using a validated screening instrument (18-item Fiber-Screen). In addition, participants reported average hours per week of leisure-time PA including club sport activities and daily screen time [mobile phone, computer and television (TV)] via a questionnaire. Results The average fiber intake per day was 14.49 g/day, less than the international recommendations of at least 25 g of fiber per day. Girls got more of their fiber from vegetables and legumes, while boys consumed more bread and potatoes. Regression analyses showed that both PA (β =0.257) and screen time, particularly computer (β =–0.144) and TV time (β =–0.138) are independently associated with fiber intake (P<0.001). Furthermore, sports club members had a significantly higher daily fiber intake than non-sports club members (16.1±6.7 vs. 12.8±4.6 g/d; P<0.001). Sports club participation was also associated with higher PA levels (40.0% vs. 15.0% reported >3 h/week) less time on the computer (51.0% vs. 33.0% reported <1 h/d) compared to non-sports club members (P<0.001) but there was no significant difference in mobile phone use and TV time. Conclusions The present results show that low fiber intake is widespread among adolescents and that longer screen time and low PA are associated with a low fiber intake, which indicates a clustering of health behaviors. Accordingly, intervention strategies may target several behaviors to promote health in youth.

Key findings

• Austrian adolescents exhibited a low daily fiber intake (mean: 14.4±6.0 g/day), with only 7.1% meeting international recommendations of 25 g/day.

• Physical activity was positively associated with daily fiber intake, whereas computer/tablet time and TV time were inversely associated. Smartphone usage was not significantly associated with fiber intake in either sex and sports club members consumed significantly more daily fiber than non-members.

• Total fiber intake did not differ by sex, but dietary fiber sources varied with girls consuming more fiber from fruits, vegetables, and legumes while fiber intake from breads and potatoes was higher among boys.

What is known and what is new?

• Inadequate dietary fiber intake, low physical activity, and excessive screen time are prevalent among adolescents globally, each posing substantial long-term metabolic, cardiovascular, and

gastrointestinal health risks.

• This study demonstrates an independent clustering between specific digital device usage, recreational physical activity and sports club participation, and daily dietary fiber intake in mid-adolescents. It highlights device- and sex-specific differences in how screen behaviors relate to dietary quality.

What is the implication, and what should change now?

• Public health interventions and pediatric guidance should avoid treating all digital media or screen time as a uniform risk factor. Furthermore, multi-component health promotion strategies should target screen modalities (TV and desktop/laptop computers) along with physical activity and leverage structured sports club engagement, and promote fiber-dense food choices to effectively improve pediatric cardiometabolic and gut health.

Introduction

The health benefits of physical activity (PA) and a balanced diet have been well documented by numerous studies (1 - 3). However, PA and nutritional behavior has changed significantly in recent decades and is often characterized by an increased level of sedentary activities (4). In particular, the increasing availability of digital media has led to an increase in screen time as television (TV), mobile phones and computers/tablets are now integral parts of daily life and shape leisure behavior, social interactions, and health-related habits (5). Various studies have shown an inverse association between screen time and PA in European adolescents (6 - 9). Objectively measured data from large cohort studies, for example, show that intensive mobile phone use is associated with less exercise and increased sedentary behavior in adolescents (10), with high screen time being considered as a cause as well as an accompanying symptom of a low level of PA and a decrease in motor performance (11). The findings of these studies are supported by current data from the World Health Organization (12 , 13), with only about 20% of children and adolescents meeting current recommendations for 60 minutes of moderate to intense PA per day (14 , 15).

Similarly, almost 80% of adolescents have been shown to exceed current screen time recommendations of2 hours per day (16 , 17). Excess screen time has also been associated with various health impairments, such as sleep disorders, depression and obesity as well as unhealthy eating habits that include high consumption of energy-rich snacks and drinks, low intake of fruit and vegetables, and irregular meals (18 - 26). Mechanistically, high screen time is hypothesized to facilitate mindless, passive eating behavior characterized by the consumption of ultra-processed convenience foods and calorie-dense snacks (27 , 28), which are inherently low in fiber. This elevated intake of low-quality foods potentially displaces or substitutes the regular consumption of nutrient-dense, high-fiber dietary options. Additionally, the environmental and situational context of screen use affects dietary habits. Screen time during family or school meals serves as a primary disruptor, which impairs cognitive awareness of food intake and dulls physiological satiety signals, driving overconsumption of low-fiber convenience items (19 , 26). Moreover, unsupervised access due to electronic media use in the bedroom not only displaces sleep but may promote late-night snacking patterns, further shifting preference away from structured, fiber-rich meals toward hyper-palatable, low-fiber alternatives (29).

Studies specifically addressing fiber intake in this context, however, remain limited. Dietary fiber is mainly found in whole grain products, fruits, vegetables, legumes and nuts and is considered a central component of a healthy diet due to positive effects on metabolic and cardiovascular health

parameters (30 , 31). Nevertheless, studies have shown that fiber intake across the world is often below the recommended levels of 25–30 g/day with 93% of Austrian university students consuming less than 25 g/d (32 - 34). Additionally, a study in Australian adolescents has shown a significant 5% decline in fiber intake between the ages of 14 and 17 years (35). The clustering of low physical activity and insufficient dietary fiber intake, therefore, represents a significant health risk during adolescence. Individually, both factors negatively impact metabolic health; however, their combined presence is strongly associated with exacerbated functional gastrointestinal disorders, such as chronic constipation, prolonged intestinal transit time, and altered gut microbiota composition (36).

From a socio-ecological perspective, health-related habits like dietary fiber intake, screen time, and PA also do not exist in a vacuum; they are heavily shaped by broader environmental and social structures. For instance, lower-income households often face barriers in obtaining fresh, high-fiber foods, and may have environmental constraints regarding safe spaces for PA that can encourage recreational screen choices (37 , 38). Furthermore, family dynamics, such as shared family meals, can facilitate structured eating routines and healthy dietary choices while limiting excessive sedentary screen habits (39).

Accordingly, the complex interactions between dietary behavior, screen time, and PA are not yet fully understood. To clarify these relationships, the present study is guided by a conceptual model, in which screen time (computer/tablet, mobile phone, TV) and PA (leisure-time PA and sports club membership) act as independent behavioral correlates of dietary fiber intake. Furthermore, sex is treated as a potential moderator of these relationships, as behavioral clustering may present differently in boys and girls (Figure 1).The overarching hypothesis is that high screen time and low PA are independently and negatively associated with fiber intake, signifying a clustering of unhealthy lifestyle behaviors. We present this article in accordance with the STROBE reporting checklist (available at https://tp.amegroups.com/article/view/10.21037/tp-2026-0432/rc).

Methods

The present study is a cross-sectional study that was carried out in the school year 2024/2025 in the federal state of Tyrol, Austria. Five schools were drawn from a random sample of the total of 28 high schools. A total of 410 adolescents between 14 and 17 years of age without known gastrointestinal conditions that could affect fiber intake and no cognitive, physical or motor impairments were included in the analyses as they provided complete data on dietary fiber intake (dependent variable) and PA as well as screen time (independent variables). This study was conducted in accordance with the Declaration of Helsinki and its subsequent amendments. The study was approved by the Ethics Committee of University of Innsbruck (No. 18/2025) and the principals of the participating schools. Parents provided written informed consent prior to data collection while participants provided oral assent.

Anthropometry

The data collection took place in the months of May and June 2025 during regular school hours in the schools' sports halls. Body height was measured with an accuracy of 0.1 cm using a mobile stadiometer (SECA ® 217, Seca, Hamburg, Germany) and body weight was measured with an accuracy of 0.1 kg using a calibrated scale (SECA ® 803, Seca, Hamburg, Germany). Body mass index (BMI) was subsequently calculated and converted to BMI percentiles using German reference values (40). These data were recorded to comprehensively characterize the study population as body composition has been shown to be associated with dietary habits, screen time and physical

activity levels in youth (41 , 42).

Fiber intake assessment

After the completion of anthropometric measurements participants completed a survey. Nutrition-related fiber intake of the adolescents was assessed using the validated 18-point Fiber-Screen questionnaire developed by Rijnaarts et al. (43). The Fiber-Screen questionnaire covers a wide range of high-fiber food categories, including whole grains, fruits, vegetables, legumes, and nuts. Participants were asked to report both the frequency (e.g., how many days per week the food is usually eaten) and the average amount per day of specific foods consumed. The participants' responses were subsequently converted to daily fiber intake (grams/day) using an evaluation algorithm that multiplies the reported frequency and amount by specific fiber content factors assigned to each food category (34 , 43).

Physical activity and screen time assessment

Additionally, participants reported their leisure time PA (average hours per week) as well as club sport membership and screen time (average hours per day) based on questions that have been used in large scale studies among German children and adolescents (44 , 45). There were no time restrictions for completing the paper-pencil questionnaire, which took around 15 minutes. Research staff was on-site to answer clarifying questions. Leisure time PA, which included recreational sports and active free play while explicitly excluding active transportation or housework, was determined using a six-point Likert scale (1= no regular PA; 6= >3 hours of leisure time PA). Club sports membership, which in Austria are organized at the community level rather than in the school system, was determined via a dichotomous question (yes/no). Even though the focus on club sports among youth is not only on competition, there are membership fees, which may limit accessibility. The average daily usage time of mobile phones, TVs and computers/tablets was recorded using a five-point Likert scale (1= <30 minutes; 5= >3 hours). Subsequently, three groups (low, moderate and high) were formed for PA and screen time based on current guidelines (46 , 47). Low: <1 h leisure time PA/week, <2 h/d mobile phone or computer/tablet time, <30 min/d TV time; moderate: 1–3 h leisure time PA/week, 2–3 h/d mobile phone or computer/tablet time, 30–60 min/d TV time; high: >3 h leisure time PA/week, >3 h/d mobile phone or computer/tablet time,>60 min/d TV time. A 3 h/week threshold for PA was applied as only recreational PA was assessed without the consideration of active transport or school-based PA, which is generally included in total PA recommendations. Confounding environmental variables, including individual socioeconomic position, the frequency of shared family meals, and total energy intake, were not collected as part of this school-based testing protocol.

Statistical analysis

Metric variables are presented as mean values with standard deviations and categorical variables as prevalence rates. Frequency differences were examined using Chi-squared tests. Due to the lack of a normal distribution, nonparametric difference tests as well as Spearman correlation analyses were used to investigate differences or relationships between fiber intake and PA as well as screen time. Group comparisons were carried out using the Kruskal Wallis test for independent samples with Bonferroni correction. In addition, a linear multiple regression was performed to evaluate associations between daily fiber intake and physical activity as well as screen time with fiber intake/day as the dependent variable and screen time/day and PA/week as independent variables. The coefficient of determination (R^2) was used as a measure for the explained variance of the regression model. In

addition to the total sample statistical analyses were performed separately for boys and girls using SPSS 30.0 (IBM, Armonk, NY, USA). Statistical significance was set at $P<0.05$, which was shifted to a statistical significance level of $P<0.01$ after applying Bonferroni correction. Figures and tables were graphically displayed using GraphPad Prism 10 (GraphPad Software, San Diego, CA, USA).

Results

A total sample of 410 participants (54.6% girls) between 14 and 17 years of age provided valid and complete data. While there was no significant sex difference in age, male participants exhibited significantly higher mean values for height, body weight, and BMI percentile compared to female participants (Table 1).

Fiber intake

Overall, the participants' average daily fiber intake was 14.49±6.01 g/day with no difference between male (14.41±6.24 g/day) and female (14.56±5.83 g/day) adolescents ($P=0.34$; Figure 2A). The lower fiber intake was also reflected by the fact that only 7.1% (n=29) of the participants met the recommended fiber intake of ≥25 g of fiber per day. Sex specific analyses revealed that 93.8% of girls (n=210) and 91.9% of boys (n=171) did not meet the current recommendations (Figure 2B). Correlation analyses did not show a significant association between age and daily fiber intake ($P=0.18$).

Most participants obtained their dietary fiber from vegetables (2.39±1.18 g), followed by legumes (1.71±2.03 g), fruits (1.36±1.01 g), and potatoes (1.35±0.87 g). The lowest fiber intake was consumed from muesli bars (0.22±0.35 g) and dried fruits (0.20±0.24 g) (Table 2).

Despite a lack of difference in total fiber intake, there were significant sex differences in fiber sources. While the fiber intake from fruits, dried fruits, vegetables, nuts and legumes was significantly higher in girls, boys consumed significantly more fiber intake from white and brown bread. For girls, the main sources of fiber were vegetables (2.52±1.24 g), legumes (1.97±2.26 g) and fruits (1.43±0.97 g), while potatoes were commonly reported sources of fiber (1.38±0.86 g) in addition to vegetables (2.23±1.10 g) and legumes among boys (1.40±1.67 g) (Table 2).

Fiber intake and screen time

More than one third of the girls (34%) and boys (37%) reported a mobile phone time of more than 3 hours per day. Within the three screen groups, the lowest usage time was observed for TV time. About 75% of the participants spent a maximum of one hour/day watching TV and a significant sex difference in screen time was only observed for computer/tablet use with boys reporting more time than girls (Table 3).

The analyses showed that the participants with low computer/tablet use (<2 h/d) consumed significantly more fiber than participants with moderate use (2–3 h/d; $P<0.05$). Similarly, a low TV time (<30 min/d) was associated with a significantly higher fiber intake compared to high TV time (>60 min/d; $P<0.01$). No differences in total fiber intake, however, were observed across mobile phone use categories (Figure 3). Similarly, correlation analyses for TV-time and computer/tablet time with daily fiber intake showed a significant inverse relationship (TV: Spearman's rho =–0.174; $P<0.001$; computer/tablet: Spearman's rho =–0.147; $P=0.003$) while no significant association was observed for mobile phone use. Furthermore, median time spent with the computer/tablet was significantly lower among participants meeting dietary fiber intake recommendations (≥25 g/d) compared to those not

meeting recommendations (30–60 vs. 1–2 h/d; P<0.001) while there were no differences for TV and smartphone time.

Fiber intake and physical activity

About half of the participants (51.2%) were members in sports clubs, with no significant difference between boys and girls. Sports club members were significantly more active than non-club sports members (median: 2–3 vs. 1.2 h/week; P<0.001). In the total sample, 38.8% had a low level of leisure time PA (<1 h/week), while just under a third (27.8%) were active for more than 3 hours per week (high). Boys reported a higher weekly PA than girls (P<0.05) (Table 4).

Daily fiber intake increased with increasing recreational PA/week (Spearman's rho =0.283; P<0.001), which resulted in a significant difference (P<0.01) in daily fiber intake between the lowest level of recreational PA (low: <1 h/week;12.9±5.6 g/d) and the highest activity level (high: >3 h/week;16.1±6.4 g/d). These differences could also be observed in the sex-specific analyses (Table 5). Daily fiber intake was also significantly higher among sports club members at 16.1±6.7 g/d than among non-sports club members at 12.8±4.6 g/d (P<0.001). Accordingly, meeting fiber intake recommendations (≥25 g/d) was associated with significantly higher PA (median: >3 h/week) compared to those not meeting recommendations (median:1–2 h/week; P<0.001) and a higher prevalence of club sports participation (89.7% vs. 48.3%; P<0.001).

Association of fiber intake with physical activity and screen time

Leisure time PA was inversely associated with TV time (Spearman rho =–0.134; P<0.01) while there were no significant associations with computer/tablet time or mobile phone use. To test the integrated conceptual model, described in the introduction, a multiple linear regression analysis was performed using daily fiber intake (g/d) as the continuous dependent variable. The independent variables entered simultaneously into the model included PA/week, computer/tablet-time/d, TV-time/d, and mobile phone-time/d as there were no concerns regarding multicollinearity (Tolerance >0.9 and VIF <1.1). Across the total sample, fiber intake was significantly associated with PA/week, computer/tablet time/d and TV-time/d (P<0.01). Specifically, higher fiber intake was associated with higher PA/week and lower computer/tablet and TV time/d, while mobile phone time/d was not significantly associated with fiber intake (P=0.80). These behaviors, however, only explained 12% of the variability of fiber intake. Sex-specific analyses showed consistent positive associations between fiber intake and PA while the inverse association with TV time was only observed in girls. Among boys, on the other hand, only computer/tablet time was inversely associated with fiber intake. As observed in the total sample the explanatory value of the variability of fiber intake, however, was low (Table 6).

Discussion

The evaluation of fiber intake among Austrian adolescents showed that (I) average fiber intake per day was below international recommendations, (II) higher PA was associated with higher fiber intake, and (III) higher computer/tablet and TV time was associated with a lower fiber intake. Sex-specific analyses further showed that particularly computer/tablet time was inversely associated with fiber intake among boys while high TV time had the strongest inverse association with fiber intake among girls.

Fiber intake in adolescents

The results showed that adolescents consume roughly 50% of the recommended 25–30 g/day of dietary fiber (30)with only 6% of girls and 8% of boys meeting current recommendations, respectively. These results are consistent with previous research in Austrian adolescents and university students as well as international youth (34 , 48 , 49). Low fiber intake among adolescents has been attributed to the high consumption of ultra-processed foods and refined carbohydrates (50 - 54) and appears to be a problem among boys and girls. The comparable results in international studies (55 - 57) suggest that a high consumption of convenience foods, which are often ultra-processed, can displace foods with a higher fiber content. In addition to total fiber, sources of fiber and quality of the overall diet should be considered as well. The present study showed that most of the dietary fiber was obtained from vegetables, legumes, fruit, potatoes and bread. Girls, however, consumed significantly more fruit, vegetables and legumes, while boys consumed potatoes and bread to a greater extent but there was no sex difference in total fiber intake.

These sex-specific differences in the consumption of certain food groups have also been observed in other studies (48 , 58). A study including more than 3600 participants found that male adolescents and adults consumed significantly more bread and fast food, while girls and women ate more fruit and vegetables (57). These fiber sourcing preferences may be driven by variations in health literacy and emerging socio-cultural gender norms during mid-adolescence (59 , 60). Prior research further notes that female adolescents often demonstrate greater health-directed motivation, with food preferences leaning toward raw fruits and vegetables, whereas male adolescents tend to prioritize portion sizes and energy density, obtaining fiber through staple starch sources like breads and processed potatoes (61). Nevertheless, these differences in food choices may not necessarily result in a sex difference in total fiber intake. The generally low fiber intake observed in the present study, however, reflects a broader global nutritional transition among adolescents. Across many countries, increasing consumption of ultra-processed foods and refined grains has displaced traditional diets rich in whole grains, legumes, fruits, and vegetables, resulting in chronically inadequate dietary fiber intake (33). Global estimates indicate that insufficient fiber consumption represents a substantial contributor to the burden of non-communicable diseases (30), emphasizing that the present findings are not unique to Austrian youth but rather mirror an international public health concern.

Fiber intake and screen time

Screen time such as time spent on mobile phones, computers and tablets play an important role in the everyday life of young people and has a strong influence on their leisure activities (5 , 8). Above all, the mobile phone is a constant companion for the majority of young people and about two-thirds of the participants in the present study reported a usage time above 2 hours/day. Nevertheless, no significant association between high mobile phone use and fiber intake was observed in the present study, while a TV-time as well as computer/tablet-time were inversely associated with daily fiber intake even though associations differed between boys and girls. A recent three-year longitudinal study with over 700 participants also indicated that prolonged screen-based sedentary behaviors were prospectively linked to declines in absolute micronutrient and fiber metrics, with distinct behavioral profiles emerging between sexes based on the specific media types consumed (21).These results suggest that screen time may affect food choices and fiber intake. Different media types, however, may affect behavioral choices differently due to differences in engagement (e.g., passive TV watching vs. interactive engagement via social media). TV viewing has historically been identified as the most harmful screen medium concerning dietary habits, largely due to its passive nature facilitating mindless snacking and its frequent exposure to unhealthy food advertising, which may lead to unhealthy food choices associated with increased consumption of high-calorie and

low-fiber foods (19 , 26 , 62). With streaming becoming more popular, a similar argument can be made for computer/tablet use as both of these media devices were inversely associated with daily fiber intake in the present study. Furthermore, social media influencers may promote unhealthy dietary choices (63). Mobile phone utilization, on the other hand, may be more fragmented and extend into time periods (such as late night in bed) when concurrent food intake is rare. Therefore, high mobile phone use may accumulate without directly affecting dietary choices.

When stratified by sex, however, TV viewing emerged as the most detrimental modality among girls, while computer/tablet time was the only significant negative correlate among boys, which may be attributed to a higher prevalence of computer gaming among boys compared to girls (64). The lack of a significant association of dietary fiber intake with mobile phones across both groups suggests that smartphone use may be more contextually fluid and less exclusively tied to stationary, home-based eating windows compared to TV sets or computers/tablets. Similarly, longitudinal evidence suggests that different screen-based behaviors (e.g., TV viewing versus electronic gaming) are associated with distinct dietary changes, and some associations differ by sex (65). Consequently, health interventions should avoid treating “screen time” as a homogeneous risk factor and instead tailor strategies to target specific devices based on the target demographic.

The previously mentioned behavioral effects may also be attributed to physiological signals of satiety and hunger triggered by excessive screen time, which can lead to unhealthy food choices associated with increased consumption of high-calorie and low-fiber foods (19 , 26 , 62). Specifically, blue light exposure has been linked to an upregulation of ghrelin while leptin may be downregulated (66). Additionally, eating while using screens may distract from physiological signals of satiety such as stomach distention and gastrointestinal hormones (28). The potential displacement of physical activity due to high screen time may further disrupt appetite regulation and elevate food reward signals that stimulate the desire for energy dense foods (67 , 68).

The interpretation of these behavioral interactions, however, should also consider underlying socioeconomic and domestic structures. Adolescents from higher socioeconomic positions generally report healthier dietary choices and are more likely to engage in structured physical activity (69), due to greater financial and structural access to sports clubs and high-fiber nutrient sources. Moreover, domestic routine acts as a powerful mediator; frequent shared family meals have been consistently linked to reduced recreational screen use during eating windows and an increased intake of vegetables, fruits, and whole grains (39).The absence of structured family meals, on the other hand, facilitates independent eating behaviors that increase the risk for mindless snacking on low-fiber convenience items during screen consumption. Therefore, the observed clustering of high screen time, low physical activity, and low fiber intake may also reflect broader socioeconomic gradients and a lack of protective family routines. Future studies, however, are needed that include objective measures of various health behaviors along with socio-economic information and longitudinal designs to enhance the understanding of the complex interactions between screen time consumption and health-related behaviors in adolescence.

Fiber intake and physical activity

While there were distinct differences in the associations between screen modalities and fiber intake between sexes (e.g., higher TV time correlated with lower fiber intake in girls vs. higher computer/tablet time correlated with lower fiber intake in boys), associations with PA were consistent for both sexes even though boys reported a higher weekly amount of PA than girls. Club sports participation was also associated with higher total PA and about half of the pupils surveyed were

members of a sports club. Nevertheless, 40% of adolescents engaged in less than 60 minutes of weekly leisure time PA, which indicates that a large proportion of adolescents does not meet current PA recommendations (70), even though other forms of PA such as active transportation or housework were not accounted for. Passive leisure time choices such as screen-based activities are often cited as the cause for the decline in PA with several studies showing an inverse association between screen time and PA as well as overall negative health effects (6 , 8 , 9 , 19). Conversely, studies show that sufficient PA and a wholesome diet are considered primary factors for good health and well-being (71 , 72).

The results of the present study indicate that higher PA is associated with a higher fiber intake. Accordingly, adolescents participating in club sports reported a significantly higher daily fiber intake than non-sports club members. Other studies also showed a positive association of exercise and sport with healthy eating habits among adolescents. Specifically, PA engagement has been associated with the consumption of higher-quality foods (e.g., whole grains, vegetables and fruit) along with beneficial effects on appetite regulation (73 - 75). The present findings are also consistent with the behavioral clustering framework, which proposes that health-related behaviors rarely occur in isolation but rather aggregate into distinct lifestyle patterns during adolescence. Previous research has demonstrated that low physical activity, prolonged recreational screen time, unhealthy dietary behaviors, and inadequate sleep frequently coexist and jointly contribute to adverse health outcomes (73 , 76 , 77). Rather than representing independent risk factors, these behaviors may reinforce one another through shared environmental, social, and psychological determinants. The concurrent association of lower fiber intake with higher recreational screen time and lower physical activity observed in the present study therefore supports the concept that dietary quality should be considered within a broader constellation of movement and lifestyle behaviors. Consequently, interventions targeting multiple behaviors simultaneously may be more effective than strategies focusing on single lifestyle components.

Strengths and limitations

Some limitations of the study should be taken into account when interpreting the findings. Data was only collected in five schools within a single federal state of Austria, which could limit generalizability of the findings. Due to the cross-sectional nature of the study, causal relationships cannot be established either. In addition, information on fiber intake, screen time and PA was obtained via self-report, which carries an inherent risk of recall bias and may lead to misclassification. Furthermore, specific screen time was assessed individually, which does not account for simultaneous use of multiple devices (e.g., mobile phone use while watching TV). It should also be considered that the 18-item fiber intake has only been validated in adults but not in adolescents. Additionally, the 18-item Fiber-Screen questionnaire does not provide a complete picture of nutrition and dietary behavior as it does not take into account total calories or other nutrients, such as fat, sugar or protein intake as well as the type of foods (e.g., home-cooked, restaurant, fast food). There was also no information on the primary purpose of screen utilization (mandatory educational use vs. recreational use) or screen use during meals as well as time of day or the physical environment in which screen media was consumed. The lack of contextual data regarding motivation for screen use and whether screens were used in environments closely linked to food availability (e.g., kitchens or living rooms) versus food-scarce settings (e.g., late-night use in bed) restricts our ability to contextualize these behavioral interfaces. Therefore, the observed associations between fiber intake and screen time as well as PA should be interpreted cautiously, as unrecorded factors such as total calorie intake, sleep or other lifestyle factors as well as environmental constraints may play a role as

well. Accordingly, the regression analyses indicated that screen time and PA account for only a minor portion of the variance in fiber intake. An important strength and novel aspect of the present study, however, is its specific focus on dietary fiber rather than overall diet quality or individual food groups. While previous studies have predominantly examined associations between screen time, physical activity, and broad dietary indices or obesity-related outcomes, relatively little attention has been paid to dietary fiber as an independent nutritional marker during adolescence. Fiber intake has been assessed with a validated questionnaire and reflects the consumption of fruits, vegetables, legumes, and whole-grain products while simultaneously serving as a proxy for overall diet quality and gastrointestinal health. Examining fiber intake, therefore, provides additional insight into the behavioral mechanisms linking recreational screen use and physical activity with dietary choices, beyond total energy intake and identifies a potentially modifiable target for future health promotion strategies.

Implications and future directions

The observed low fiber intake emphasizes the need for better nutrition education, whereby the school could be an ideal setting, because it reaches most children and adolescents, regardless of their socio-economic background. Interventions may include healthier meal choices in schools and potentially limiting the access of unhealthy choices in vending machines. In addition, club sports may serve as viable intervention setting given that nutrition is also vital for physical fitness and sports performance. However, the generally weak associations demonstrate that the relationship between recreational choices and fiber intake is complex. This was also reflected by the low explanatory values of the multivariate regression, which emphasizes the need for more encompassing health promotion strategies. A better understanding of relationships between behavioral choices, therefore, is needed to develop targeted interventions. Accordingly, future studies should adopt a more holistic, socio-ecological approach by incorporating objective metrics of socioeconomic status, documenting the home food environment (including family meal structures), and capturing broader dietary behaviors. This will allow multivariate models to pinpoint whether screen-media interventions, parental role modeling, or socio-economic safety nets offer the highest efficacy for behavioral change.

Conclusions

This study highlights a significant deficiency in the fiber intake of adolescents even though it is considered a central component of a healthy diet. In addition, fiber consumption was associated with PA and screen time. Active students and sports club members consumed more fiber than inactive students and non-sports club members, while high screen time, particularly, TV and computer/tablet time was associated with a lower daily fiber intake. The associations between various screen-based recreational choices and fiber intake, however, differ between boys and girls. These results are consistent with previous research and indicate that various health behaviors may cluster among adolescents.

Supplementary